

MELBOURNE VIEWBANK, AustraliaWMO: **958740**Lat: **37.7408S**Lon: **145.0972E**Elev: **66**StdP: **100.53**Time Zone: **10.00 (AUS)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	3.3	0.2	3.9	11.6	1.2	4.1	10.5	9.6	13.3	8.8	12.6	1.6	40	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	35.4	19.3	32.9	18.9	30.6	18.4	21.6	28.7	20.5	28.1	19.7	27.0	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	22.4	18.6	13.6	21.9	17.6	12.7	21.5	63.1	28.8	59.4	28.5	56.5	27.0	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.1	7.1	-0.4	40.7	0.8	2.4	-1.0	42.4	-1.5	43.9	-1.9	45.2	-2.5	47.0		
			-0.5	23.3	0.8	1.1	-1.1	24.1	-1.6	24.8	-2.0	25.4	-2.6	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	20.8	20.4	18.9	15.7	12.6	10.2	10.0	10.5	12.3	14.4	17.0	18.7
	DBStd	4.95	4.04	3.53	3.53	2.98	2.51	2.10	2.02	2.26	2.72	3.25	3.83	3.82
	HDD10.0	79	0	0	0	1	5	22	26	20	5	1	0	0
	HDD18.3	1506	16	16	36	91	179	243	259	242	182	129	72	41
	CDD10.0	1938	334	291	276	170	84	29	25	36	75	138	209	271
	CDD18.3	324	92	74	53	11	1	0	0	0	2	8	31	53
	CDH23.3	3759	1070	755	541	112	2	0	0	0	13	154	425	688
	CDH26.7	1654	550	335	217	19	0	0	0	1	42	167	323	
Wind	WSAvg	3.2	3.5	3.2	3.0	2.7	2.9	2.7	3.2	3.2	3.3	3.4	3.4	3.4
Precipitation	PrecAvg	766	46	59	45	67	63	66	68	67	78	68	74	65
	PrecMax	1085	109	182	92	160	143	137	109	112	151	159	167	171
	PrecMin	479	0	5	14	20	16	19	26	25	24	12	28	8
	PrecStd	155	25	44	22	36	29	26	23	26	33	37	35	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	36.6	35.4	29.0	22.9	18.1	17.4	20.2	25.0	30.7	34.0	38.0
		MCWB	20.4	19.6	19.3	16.8	14.3	12.7	11.6	12.7	14.8	16.7	18.5	18.7
	2%	DB	35.8	33.5	31.8	26.4	20.4	16.2	15.7	18.2	21.9	27.3	31.2	33.8
		MCWB	19.5	19.4	18.4	16.1	13.9	11.9	11.1	11.7	13.9	15.8	17.6	18.7
	5%	DB	32.7	30.9	28.9	24.2	18.9	15.1	14.6	16.4	19.9	24.6	28.3	30.2
		MCWB	19.3	19.1	17.9	15.6	13.4	11.5	10.5	11.0	12.8	15.0	17.4	18.0
	10%	DB	29.3	28.2	26.2	22.1	17.5	14.0	13.6	14.9	18.1	22.0	25.3	27.1
		MCWB	18.7	18.7	17.3	14.9	12.8	11.0	10.0	10.4	11.7	14.0	16.5	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.6	21.2	18.3	16.4	13.9	13.2	13.5	15.8	18.3	20.9	21.7
		MCDB	30.3	29.9	28.4	24.7	19.4	15.8	15.5	18.2	21.8	24.8	28.1	28.5
	2%	WB	21.8	21.2	20.0	17.3	14.8	12.8	12.0	12.4	14.5	16.9	19.4	20.1
		MCDB	28.8	28.6	25.6	22.7	18.5	14.8	14.3	16.6	20.3	24.3	26.4	28.2
	5%	WB	20.7	20.2	19.2	16.5	13.9	12.1	11.2	11.5	13.5	15.7	18.4	19.1
		MCDB	29.1	27.3	25.3	21.7	17.8	14.3	13.4	15.2	18.7	22.5	25.3	26.8
	10%	WB	19.6	19.3	18.3	15.7	13.2	11.4	10.5	10.8	12.5	14.6	17.2	18.2
		MCDB	27.8	26.1	24.8	20.7	16.6	13.6	12.8	14.1	16.9	20.9	23.9	25.6
Mean Daily Temperature Range	5% DB	MDBR	12.1	11.5	11.0	10.3	8.0	7.0	6.6	7.6	9.1	11.0	11.4	12.0
		MCDBR	19.0	17.4	16.6	14.6	10.5	8.9	7.7	9.8	12.6	16.8	17.7	18.9
		MCWBR	5.6	5.5	5.6	6.0	5.5	5.5	4.5	5.1	6.1	7.1	6.3	6.0
	5% WB	MCDBR	14.6	13.7	12.7	11.9	8.7	7.5	6.7	8.6	10.9	14.0	14.1	14.5
		MCWBR	5.4	5.5	5.0	5.4	5.1	5.0	4.4	4.9	5.7	6.6	5.8	5.8
Clear-Sky Solar Irradiance	taub	0.350	0.356	0.332	0.330	0.310	0.302	0.302	0.313	0.336	0.337	0.347	0.340	
	taud	2.509	2.499	2.564	2.540	2.582	2.591	2.582	2.542	2.469	2.491	2.489	2.528	
	Ebn at Noon	983	953	938	875	836	814	836	882	916	962	979	997	
	Edh at Noon	112	108	94	84	70	64	69	82	101	108	113	111	
All-Sky Solar Radiation	RadAvg	7.16	6.25	4.99	3.46	2.27	1.76	1.96	2.78	3.97	5.33	6.29	7.04	
	RadStd	0.45	0.33	0.28	0.21	0.14	0.15	0.11	0.20	0.29	0.44	0.38	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.52	N/A	N/A	N/A	+0.85	+1.48	N/A	-103	+210	+113	(46)	
(47)	Regional (40 neighbors)	+0.51	N/A	N/A	+0.85	N/A	N/A	-18	-127	+173	+59	(47)	

Nomenclature: See separate page